PROBLEM 50

Double Pendulum

Original statement · Problems.pdf, p. 35

Kinematic Linkage

A double pendulum consists of two heavy point masses P and Q, each of mass m, connected to the origin O and to each other by two rigid, massless bars of length ℓ as shown in the figure. We define the inertial frame S_0 with origin O and vector basis $\mathcal{B}_0$. We define the two angles θ and ϕ shown in the figure, to describe the configuration of the system.

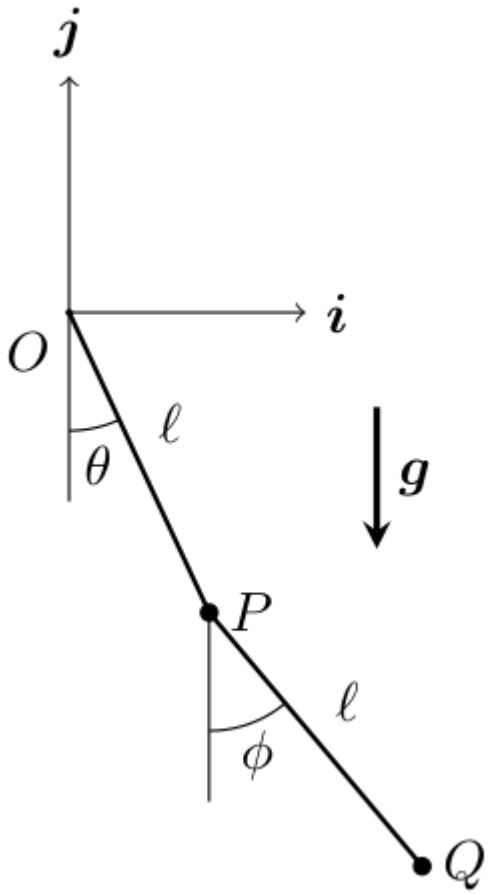


Figure: Double pendulum O–P–Q with bars of length ℓ, angle θ of OP and angle φ of PQ from the vertical (Problems.pdf, p. 35).

(a) Write down the general expressions of the position, velocity, and acceleration vectors of P and Q, using θ, ϕ and their [derivatives].

(b) Write down all forces acting on P and Q, introducing new unknowns as needed.

(c) Write down the equations of motion for θ and ϕ, free of other unknowns.

(d) Find and write down one conservation law of this problem.

(e) If the initial conditions are $\theta = \phi = \pi/4, \dot{\theta} = \dot{\phi} = 0$, will Q ever cross the Ox axis? Justify your answer.

(f) Consider now that we impose the additional constraint $\phi = 2\theta$. Compute the natural frequency of oscillation of the system under the assumption of small angles.

(Hint: The mechanical energy of an ideal harmonic oscillator of natural frequency ω_0 and mass m can be written as $\frac{1}{2}m(\dot{x}^2 + \omega_0^2 x^2)$).

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a double pendulum of two heavy particles P and Q of mass m joined to O and to each other by two rigid massless bars of length ℓ. The motion is planar (the figure and the angles θ and ϕ describe a plane). The angle θ is that of the bar OP with the downward vertical and ϕ that of the bar PQ with the downward vertical.
- **Constraints:** two particles in the plane have 4 coordinates; the rigid bars give $|OP| = \ell$ and $|PQ| = \ell$ (holonomic, scleronomic, bilateral). $\mathrm{CDOF} = 4 - 2 = 2$, with (θ, ϕ).
- **Parameters:** m, ℓ, g.

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$, with $\mathbf{i}$ horizontal to the right and $\mathbf{j}$ vertical upward (the horizontal axis is Ox).
- **Bar bases** (rotations about the axis perpendicular to the plane; check below):

$$\mathbf{e}_P = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}, \quad \mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}, \qquad \mathbf{e}_Q = \sin\phi\,\mathbf{i} - \cos\phi\,\mathbf{j}, \quad \mathbf{e}_\phi = \cos\phi\,\mathbf{i} + \sin\phi\,\mathbf{j}$$

In $\{\mathbf{i}, \mathbf{j}\}$ the matrix with columns $\mathbf{e}_P, \mathbf{e}_\theta$ is $\begin{pmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{pmatrix}$, with $\det = \sin^2\theta + \cos^2\theta = 1$ and orthonormal columns ($\sin^2\theta + \cos^2\theta = 1, \sin\theta\cos\theta - \cos\theta\sin\theta = 0$), so $RR^T = I$; the same holds with ϕ for $\mathbf{e}_Q, \mathbf{e}_\phi$. where $\mathbf{e}_P$ points from O to P and $\mathbf{e}_Q$ from P to Q. Their derivatives are $\dot{\mathbf{e}}_P = \dot\theta\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_P$, and the same with ϕ.

- **Scalar products needed:** $\mathbf{j}\cdot\mathbf{e}_P = -\cos\theta, \mathbf{j}\cdot\mathbf{e}_\theta = \sin\theta, \mathbf{j}\cdot\mathbf{e}_Q = -\cos\phi, \mathbf{j}\cdot\mathbf{e}_\phi = \sin\phi$, and

$$\mathbf{e}_P\cdot\mathbf{e}_Q = \cos(\theta-\phi), \quad \mathbf{e}_\theta\cdot\mathbf{e}_Q = \sin(\phi-\theta), \quad \mathbf{e}_\theta\cdot\mathbf{e}_\phi = \cos(\theta-\phi), \quad \mathbf{e}_P\cdot\mathbf{e}_\phi = \sin(\theta-\phi)$$

Derivation

3.1 Kinematics

(a) Position, velocity and acceleration

> **Pedagogical Justification:** the positions are sums of the two bar vectors, $\mathbf{r}_P = \ell\,\mathbf{e}_P$ and $\mathbf{r}_Q = \ell\,\mathbf{e}_P + \ell\,\mathbf{e}_Q$; the derivatives use Poisson's formula:

$$\mathbf{r}_P = \ell\sin\theta\,\mathbf{i} - \ell\cos\theta\,\mathbf{j}, \qquad \mathbf{r}_Q = \ell\,(\sin\theta + \sin\phi)\,\mathbf{i} - \ell\,(\cos\theta + \cos\phi)\,\mathbf{j}$$

$$\mathbf{v}_P = \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{v}_Q = \ell\dot\theta\,\mathbf{e}_\theta + \ell\dot\phi\,\mathbf{e}_\phi, \qquad v_P^2 = \ell^2\dot\theta^2, \quad v_Q^2 = \ell^2\left[\dot\theta^2 + \dot\phi^2 + 2\dot\theta\dot\phi\cos(\theta-\phi)\right]$$

$$\mathbf{a}_P = \ell\left(\ddot{\theta}\,\mathbf{e}_\theta - \dot{\theta}^2\,\mathbf{e}_P\right), \qquad \mathbf{a}_Q = \mathbf{a}_P + \ell\left(\ddot{\phi}\,\mathbf{e}_\phi - \dot{\phi}^2\,\mathbf{e}_Q\right)$$

3.2 Forces

(b) Forces on P and Q

- The bars are massless and rigid, so each transmits a force along its axis: T_1 in OP and T_2 in PQ (negative values mean compression).
- **On P:** its weight $-mg\,\mathbf{j}$, the pull $-T_1\,\mathbf{e}_P$ of the bar OP and the pull $+T_2\,\mathbf{e}_Q$ of the bar PQ. **On Q:** its weight $-mg\,\mathbf{j}$ and the pull $-T_2\,\mathbf{e}_Q$ of the bar PQ (action and reaction). **Inertia forces:** none.

3.3 Equations of Motion

(c) Equations of motion for θ and ϕ

> **Pedagogical Justification:** Newton's second law, $m\mathbf{a}_P = -mg\,\mathbf{j} - T_1\mathbf{e}_P + T_2\mathbf{e}_Q$ and $m\mathbf{a}_Q = -mg\,\mathbf{j} - T_2\mathbf{e}_Q$, gives four scalar equations for the four unknowns θ, ϕ, T_1, T_2. The two tensions are removed by choosing the projection directions.

- **Equation for ϕ.** Project the equation of Q on $\mathbf{e}_\phi$, which is orthogonal to $\mathbf{e}_Q$ and removes T_2. With $\mathbf{a}_P \cdot \mathbf{e}_\phi = \ell\left[\ddot{\theta}\cos(\theta-\phi) - \dot{\theta}^2\sin(\theta-\phi)\right]$ and $\ell\left(\ddot{\phi}\,\mathbf{e}_\phi - \dot{\phi}^2\mathbf{e}_Q\right)\cdot\mathbf{e}_\phi = \ell\ddot{\phi}$:

$$\boxed{\ddot{\phi} + \ddot{\theta}\cos(\theta-\phi) - \dot{\theta}^2\sin(\theta-\phi) + \frac{g}{\ell}\sin\phi = 0}$$

- **Equation for θ.** Add the two equations of motion so that $T_2\,\mathbf{e}_Q$ cancels, and project on $\mathbf{e}_\theta$, which is orthogonal to $\mathbf{e}_P$ and removes T_1. The right-hand side is $-2mg\,\mathbf{j}\cdot\mathbf{e}_\theta = -2mg\sin\theta$, and the left-hand side is $m\left(\mathbf{a}_P + \mathbf{a}_Q\right)\cdot\mathbf{e}_\theta$ with $\mathbf{a}_P\cdot\mathbf{e}_\theta = \ell\ddot{\theta}$ and $\mathbf{a}_Q\cdot\mathbf{e}_\theta = \ell\ddot{\theta} + \ell\ddot{\phi}\cos(\theta-\phi) + \ell\dot{\phi}^2\sin(\theta-\phi)$:

$$\boxed{2\ddot{\theta} + \ddot{\phi}\cos(\theta-\phi) + \dot{\phi}^2\sin(\theta-\phi) + \frac{2g}{\ell}\sin\theta = 0}$$

- **Tensions (afterwards).** Projecting on the bar directions,

$$T_2 = mg\cos\phi - m\ell\left[\ddot\theta\sin(\phi-\theta) - \dot\theta^2\cos(\theta-\phi) - \dot\phi^2\right], \qquad T_1 = mg\cos\theta + m\ell\dot\theta^2 + T_2\cos(\theta-\phi)$$

At rest at $\theta = \phi = 0$: $T_2 = mg$ and $T_1 = 2mg$, the static tensions of the hanging chain.

3.4 Integration and Answers

(d) Conservation law

> **Pedagogical Justification:** the bar forces do no net work. Bar OP acts on P with $-T_1\mathbf{e}_P$, whose power is $-T_1\mathbf{e}_P\cdot\ell\dot\theta\,\mathbf{e}_\theta = 0$. Bar PQ acts with $+T_2\mathbf{e}_Q$ on P and $-T_2\mathbf{e}_Q$ on Q, total power $-T_2\mathbf{e}_Q\cdot(\mathbf{v}_Q - \mathbf{v}_P) = -T_2\mathbf{e}_Q\cdot\ell\dot\phi\,\mathbf{e}_\phi = 0$. Gravity is conservative, so the mechanical energy is constant:

$$E = \tfrac{1}{2}m\ell^2\left[2\dot\theta^2 + \dot\phi^2 + 2\dot\theta\dot\phi\cos(\theta-\phi)\right] - mg\ell\,(2\cos\theta + \cos\phi) = \text{const}$$

(e) Will Q ever cross the Ox axis?

> **Pedagogical Justification:** with $\theta(0) = \phi(0) = \pi/4$ and $\dot\theta(0) = \dot\phi(0) = 0$, the initial energy is purely potential:

$$E_0 = -mg\ell\left(2\cos\frac{\pi}{4} + \cos\frac{\pi}{4}\right) = -\frac{3\sqrt{2}}{2}mg\ell \approx -2.121\,mg\ell$$

- Q is on the axis when $y_Q = -\ell\,(\cos\theta + \cos\phi) = 0$, i.e. $\cos\phi = -\cos\theta$. There the potential energy is $V = -mg\ell\,(2\cos\theta + \cos\phi) = -mg\ell\cos\theta \geq -mg\ell$. Since the kinetic energy is non-negative, $E \geq V$ at any point of the motion, so reaching the axis would require $E_0 \geq -mg\ell$. But $E_0 = -2.121\,mg\ell < -mg\ell$.
- **Answer:** Q can **never** cross the Ox axis (it would need a negative kinetic energy).

(f) Additional constraint $\phi = 2\theta$: small-angle natural frequency

> **Pedagogical Justification:** with the constraint $\phi = 2\theta$ the system has one degree of freedom. The additional ideal constraint (scleronomic, no friction) does no work, so the energy is still conserved. For small angles, $\dot\phi = 2\dot\theta$, $\cos(\theta-\phi) = \cos\theta \approx 1$ and $\cos u \approx 1 - u^2/2$:

$$K = \tfrac{1}{2} m\ell^2 \left[2 + 4 + 4\right] \dot\theta^2 = 5 m\ell^2 \dot\theta^2, \qquad V = -mg\ell \left[2\left(1 - \tfrac{\theta^2}{2}\right) + \left(1 - 2\theta^2\right)\right] = -3mg\ell + 3mg\ell\theta^2$$

$$E = 5m\ell^2\dot\theta^2 + 3mg\ell\theta^2 + \text{const} \implies E' = 10m\ell^2\dot\theta\ddot\theta + 6mg\ell\theta\dot\theta = 0 \implies \ddot\theta + \frac{3g}{5\ell}\theta = 0$$

- **Comparison with the hint.** The energy of an ideal oscillator is $\tfrac{1}{2} m_{\text{eq}} \left(\dot x^2 + \omega_0^2 x^2\right)$; here $\tfrac{1}{2}(10m\ell^2)\left(\dot\theta^2 + \dfrac{3g}{5\ell}\theta^2\right)$, so

$$\boxed{\omega_0 = \sqrt{\frac{3g}{5\ell}}}$$

Interpretation & Checks

- **Dimensional check:** $[g/\ell] = \mathrm{s}^{-2}$, so $[\omega_0] = \mathrm{s}^{-1}$; $[T_1] = [T_2] = [mg] = \mathrm{N}$.
- **Comparison (not a limit of this system):** for a single pendulum the frequency would be $\sqrt{g/\ell}$; the constrained double pendulum oscillates more slowly ($\omega_0^2 = 0.6\, g/\ell$) because the second mass adds inertia faster than restoring torque.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 50 in its solution keys, so the results are verified by independent derivation. Sympy confirms that the Euler-Lagrange equations of $K - V$ are $m\ell^2$ times the two equations of (c), and gives $K \approx 5m\ell^2\dot\theta^2$ and $V''(0) = 6mg\ell$ for the constrained case.

Source: Problems.pdf